No of Pages : 4 **Course Code : 23MX12**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, MARCH 2025

MCA Semester : 1

23MX12 STRUCTURED PROGRAMMING CONCEPT

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivisions (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.

1. a) Compare and contrast top down approach and bottom approach for structured programming.

b) i) Discuss on various criteria for classification of programming languages. Based on the secriteria, fit the position of C Programming language.? (5)

ii) Consider the following program. Write the output for every input shown in the table (5)

```
#include<stdio.h>
#include<conio.h>
int main() {
int x=0,n;
clrscr();
scanf("%d",&n);
if(n>8) {
if(n<9)   x=1; }
else {
if(n>4)
if(n<6) x=2;
else x=3;
else
 if(n>2)  x=4;
 else
  x=5;   }
printf("%d",x);   }
```

Input	1	3	7	9
Output				

c) Analyze the properties of good programming language in the perspective of programmer and language developer. Enumerate the reasons for the popularity of a programming language. Give simple examples in C programming for the above points.

2. a) Demonstrate a situation where switch case is better than if else ladder. Find the output of the program

```
#include<stdio.h>
#include<conio.h>
void main()  {
        clrscr();
        switch(printf("a"),printf("abc"),printf("ab"))
        {
        case 1: switch(printf("first"))
        break;
        case 2: switch(printf("second"))
         break;
        default : switch(printf("third"))
         break;
        }       }
```

b) Analyze following aspects of structured programming language with C language and compare them.

i) Scope and lifetime of variables.

ii) Call by value and call by address

iii) Iterative structure statements and its sequence control

c) i) 1) Explain the various components of user defined function with example? Also distinguish iteration from recursion in software application development with suitable illustration. (8)

2) Develop a C function to perform the following task. Ramya has two numbers, her tuition teacher give a task to her. The task is that, she wants to find the sum of all digits appearing between those two numbers. (4)

Num1 = 8 and Num2 = 13

Output: 27

Explanation: (8 + 9 + 1 + 0 + 1 + 1 + 1 + 2 + 1 +3)

Constraints

0 <= Number1 <=10000

Number1 <= Number2 <=10000

(OR)

ii) Consider the C program below (with gaps to be filled in) to perform the following task:
Reads a positive integer *k* (0 < *k* <5) and then prints all permutations (with repetitions allowed) of length *k* using the digits from 1 to 9. Note that since repetitions are allowed, there will be 9*k* permutations for a given *k*. You are required to fill up the missing lines in the program.

Note*: Interchanging* i2, i3, *and* i4 *as loop control is okay as long as the correct permutations are printed*

```
void main()
{
int k, count = 1, i1, i2, i3, i4;
scanf("%d", &k);
printf("k = %d\n", k);
for (i1 = 1; i1 < 10; ++i1)
if (k == 1) {
printf("Permutation No %d = ", count++);
printf("%d\n", i1);
}                                                    <-----
else
for (i2 =-----------------------)
if (k == 2) {
printf("Permutation No %d = ", count++);
 printf("   %d, %d\n", ----------------
 );                                                  <-----
}
else
for (i3----------------------)                       <-----

if (k==3) {
printf("Permutation No %d = ", count++);
 printf---------------------------",);
                                                     <-----
}
else
 for (i4 ----------------------)
 if (k==4) {                                         <-----
 ------                                              <-----
printf("Permutation No %d = ", count++);
printf--------------------------------);

}                                                    <-----
}
```

3. a) Differentiate constant pointer and a pointer to a constant.

b) Explain any five string functions used in C?
Write a C program to implement the following. Given a string s, you have to replace all instances of
i. ‘AND’ with ‘&’
ii. ‘QN’ with ‘?’
and print the resultant string.
Test case:
Consider s = **‘and_is_AND_but_and_is_not_AND_though_QN_is_?’**
The output must be
‘and_is_&_but_and_is_not_&_though_?_is_?’

c) i) 1) Differentiate static and dynamic memory allocation and Discuss the functions used in dynamic memory allocation. (6)

2) Implement using dynamic memory allocation, pointer and pointer arithmetic operations to accept two matrices and store them in column major form and perform matrix concatenation of corresponding matrix elements.

X[i][j] =3 and Y[i][j]=7 and the result Z[i][j] =37 (6)

(OR)

ii) Write a structure to store the name, account number and balance of customers (more than 5) and store their information in a file.

1) Write a C program that opens the file and appends the given new customer records to the file.

2) Display the content of the file.

3) Write a function to print the names of all the customers having balance less than Rs.200.

4. a) Under what scenario, scripting languages are used for developing applications.

b) Compare multi-paradigm languages with domain-specific languages, and also discuss when would you choose one over the other with typical use cases?

c) Explain the concept of static and dynamic libraries using C programming.

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